

MariaDB_dataset-2

Commit 21:

5f7c2a617f4712c2f698ea7f98eb19de68bae277

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, p.title, c.comment FROM users u, posts p, comments c
WHERE u.id = p.user_id AND p.id = c.post_id ORDER BY p.title DESC;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  4.3572000000000001
4  The new version is executed by default
5  4.3676
6  The old version
7  4.3846
8  Comparison of New and old versions (along symbolic execution path)
9  -0.00624914473384099
10 Comparison of New and old versions (along the default execution path)
11 -0.003877206586689655
```

No performance regression

The 2nd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
```

```
2 The new version executes along symbols
3 0.3356
4 The new version is executed by default
5 0.333999999999999996
6 The old version
7 0.3383
8 Comparison of New and old versions (along symbolic execution path)
9 -0.007981081879988118
10 Comparison of New and old versions (along the default execution path)
11 -0.01271061188294421
```

No performance regression

The 3rd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3389
4 The new version is executed by default
5 0.3384
6 The old version
7 0.3355
8 Comparison of New and old versions (along symbolic execution path)
9 0.010134128166914929
10 Comparison of New and old versions (along the default execution path)
11 0.008643815201192125
```

No performance regression

The 4th symbolic execution

Testcase:SELECT * FROM products WHERE FIND_IN_SET('New', tags);

No conditional branches, no need for symbolic execution.

代码块
end-to-end diff:

No performance regression

The 5th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

1 end-to-end diff:

No performance regression

The 6th symbolic execution

Testcase:SELECT username FROM users WHERE id BETWEEN 700 AND 1200 ORDER BY username;

No conditional branches, no need for symbolic execution.

代码块

1 end-to-end diff:

No performance regression

The 7th symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

1 end-to-end diff:
2 The new version executes along symbols
3 13.689599999999999

```
4 The new version is executed by default
5 13.857099999999999
6 The old version
7 14.073699999999999
8 Comparison of New and old versions (along symbolic execution path)
9 -0.02729204118319988
10 Comparison of New and old versions (along the default execution path)
11 -0.015390409060872386
```

No performance regression

The 8th symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution exception occurred, resulting in a timeout.

old异常

No performance regression

Commit 22:

ab468e33aff110b44a31ce0350894906ff4bc757

Non-C file modifications, therefore no performance regression testing is required.

Commit 23:

3b3c512febce627d5125d542754cdc9fbc2164f1

Unable to provide test cases

Commit 24:

22efc2c784e1b7199fb5804e6330168277ea7dce

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.346
4  The new version is executed by default
5  0.3442
6  The old version
7  0.3472
8  Comparison of New and old versions (along symbolic execution path)
9  -0.003456221198156781
10 Comparison of New and old versions (along the default execution path)
11 -0.008640552995391713
```

No performance regression

The 2nd symbolic execution

Testcase:SELECT DISTINCT u.username, p.title, c.comment FROM users u, posts p, comments c
WHERE u.id = p.user_id AND p.id = c.post_id ORDER BY p.title DESC;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
```

No performance regression

The 3rd symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  13.714699999999999
4  The new version is executed by default
5  13.5116000000000001
6  The old version
```

```
7 13.9218
8 Comparison of New and old versions (along symbolic execution path)
9 -0.014875949949000884
10 Comparison of New and old versions (along the default execution path)
11 -0.02946458072950322
```

No performance regression

The 4th symbolic execution

Testcase:DELETE IGNORE FROM comments WHERE id = 1;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.0032
4 The new version is executed by default
5 0.0031
6 The old version
7 0.0032
8 Comparison of New and old versions (along symbolic execution path)
9 0.0
10 Comparison of New and old versions (along the default execution path)
11 -0.031250000000000008
```

No performance regression

The 5th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 6th symbolic execution

Testcase:CALL GetUserPosts(1);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0002
4  The new version is executed by default
5  0.0002
6  The old version
7  0.0002
8  Comparison of New and old versions (along symbolic execution path)
9  0.0
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 7th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  3.3627
4  The new version is executed by default
5  3.3890
6  The old version
7  3.3751
8  Comparison of New and old versions (along symbolic execution path)
9  -0.00367396522
10 Comparison of New and old versions (along the default execution path)
```

```
11 0.00411839649
```

No performance regression

The 8th symbolic execution

Testcase:SELECT * FROM posts WHERE MATCH(title, content) AGAINST('search term' IN NATURAL LANGUAGE MODE);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0418
4  The new version is executed by default
5  0.0427
6  The old version
7  0.0459
8  Comparison of New and old versions (along symbolic execution path)
9  -0.08932461873638357
10 Comparison of New and old versions (along the default execution path)
11 -0.06971677559912856
```

No performance regression

The 9th symbolic execution

Testcase:DESC users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0138000000000000002
4  The new version is executed by default
5  0.0108
6  The old version
7  0.0145999999999999998
8  Comparison of New and old versions (along symbolic execution path)
9  -0.054794520547945
```

```
10 Comparison of New and old versions (along the default execution path)
11 -0.2602739726027396
```

No performance regression

The 10th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id=1000;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 11th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id=1000;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.107400000000000001
4 The new version is executed by default
5 0.123900000000000001
6 The old version
7 0.11779999999999999
8 Comparison of New and old versions (along symbolic execution path)
9 -0.08828522920203719
10 Comparison of New and old versions (along the default execution path)
11 0.05178268251273364
```

No performance regression

The 12th symbolic execution

Testcase:SELECT * FROM posts WHERE MATCH(title, content) AGAINST('search term' IN NATURAL LANGUAGE MODE);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0486
4  The new version is executed by default
5  0.0465
6  The old version
7  0.0456
8  Comparison of New and old versions (along symbolic execution path)
9  0.06578947368421043
10 Comparison of New and old versions (along the default execution path)
11 0.019736842105263115
```

No performance regression

The 13th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0027
4  The new version is executed by default
5  0.0038
6  The old version
7  0.0038
8  Comparison of New and old versions (along symbolic execution path)
9  -0.28947368421052627
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 14th symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  13.6501
4  The new version is executed by default
5  13.8814
6  The old version
7  14.2543
8  Comparison of New and old versions (along symbolic execution path)
9  -0.04238720947363255
10 Comparison of New and old versions (along the default execution path)
11 -0.026160526998870608
```

No performance regression

The 15th symbolic execution

Testcase:SELECT * FROM posts WHERE MATCH(title, content) AGAINST('search term' IN NATURAL LANGUAGE MODE);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0429
4  The new version is executed by default
5  0.042499999999999996
6  The old version
7  0.056200000000000001
8  Comparison of New and old versions (along symbolic execution path)
9  -0.2366548042704627
10 Comparison of New and old versions (along the default execution path)
11 -0.24377224199288272
```

No performance regression

The 16th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.33999999999999997
4  The new version is executed by default
5  0.3352
6  The old version
7  0.3355
8  Comparison of New and old versions (along symbolic execution path)
9  0.013412816691505063
10 Comparison of New and old versions (along the default execution path)
11 -0.000894187779433748
```

No performance regression

The 17th symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  13.6376
4  The new version is executed by default
5  13.6069
6  The old version
7  13.7681
8  Comparison of New and old versions (along symbolic execution path)
9  -0.009478432027658109
10 Comparison of New and old versions (along the default execution path)
```

```
11  -0.011708224083206898
```

No performance regression

The 18th symbolic execution

Testcase:SELECT * FROM posts WHERE MATCH(title, content) AGAINST('search term' IN NATURAL LANGUAGE MODE);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0376
4  The new version is executed by default
5  0.0455
6  The old version
7  0.0469
8  Comparison of New and old versions (along symbolic execution path)
9  -0.1982942430703624
10 Comparison of New and old versions (along the default execution path)
11 -0.029850746268656685
```

No performance regression

The 19th symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  14.2497
4  The new version is executed by default
5  14.127699999999999999
6  The old version
7  14.284600000000000001
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0024431905688643975
```

```
10 Comparison of New and old versions (along the default execution path)
11 -0.010983856740825927
```

No performance regression

The 20th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users (id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT NULL, email VARCHAR(100) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.002900000000000000002
4 The new version is executed by default
5 0.002900000000000000002
6 The old version
7 0.002900000000000000002
8 Comparison of New and old versions (along symbolic execution path)
9 0.0
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 21st symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 14.0765
4 The new version is executed by default
5 13.9184
6 The old version
```

```
7 14.2095
8 Comparison of New and old versions (along symbolic execution path)
9 -0.009359935254583264
10 Comparison of New and old versions (along the default execution path)
11 -0.020486294380520083
```

No performance regression

The 22nd symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 23rd symbolic execution

Testcase:UPDATE employee SET age = age + 1 WHERE age < 30;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

Commit 25: cb2d6abae1996d62bbf5f3e2f4594010546add3b

The 1st symbolic execution

Testcase:SELECT u.username, COUNT(p.id) AS post_count, MAX(p.created_at) AS last_post_time
FROM users u LEFT JOIN posts p ON u.id = p.user_id GROUP BY u.username HAVING COUNT(p.id)
> 1 AND MAX(p.created_at) > '2025-01-01';

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  75.85159999999999
4  The new version is executed by default
5  75.5958
6  The old version
7  75.6152
8  Comparison of New and old versions (along symbolic execution path)
9  0.003126355547561721
10 Comparison of New and old versions (along the default execution path)
11 -0.0002565621726849168
```

No performance regression

Commit 26:

f02ad2f641dc0ff8c6bf93cc829b2ca40c7e8cad

Non-C file modifications, therefore no performance regression testing is required.

Commit 27:

c36cd5604923545aa0583de1958c4d114911326a

Non-C file modifications, therefore no performance regression testing is required.

Commit 28:

1e00b9ec2a13a3d7039fd4b9803a6a522c161a81

Unable to provide test cases

Commit 29:

f11504af51535bb2f5035597afa61f79012da05e

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY

u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3386
4  The new version is executed by default
5  0.3383
6  The old version
7  0.33599999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  0.007738095238095378
10 Comparison of New and old versions (along the default execution path)
11 0.006845238095238168
```

No performance regression

The 2nd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3376
4  The new version is executed by default
5  0.3376
6  The old version
7  0.3321
8  Comparison of New and old versions (along symbolic execution path)
9  0.016561276723878363
10 Comparison of New and old versions (along the default execution path)
11 0.016561276723878363
```

No performance regression

The 3rd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-01') GROUP BY u.username HAVING COUNT(p.id) > 1;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.7841
4  The new version is executed by default
5  0.785700000000000001
6  The old version
7  0.778899999999999999
8  Comparison of New and old versions (along symbolic execution path)
9  0.006676081653614192
10 Comparison of New and old versions (along the default execution path)
11 0.008730260623957042
```

No performance regression

The 4th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3369000000000000003
4  The new version is executed by default
5  0.3384999999999999997
6  The old version
7  0.3368
8  Comparison of New and old versions (along symbolic execution path)
9  0.0002969121140143839
10 Comparison of New and old versions (along the default execution path)
```

11 0.005047505938242219

No performance regression

The 5th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.337
4  The new version is executed by default
5  0.3377
6  The old version
7  0.3342
8  Comparison of New and old versions (along symbolic execution path)
9  0.008378216636744538
10 Comparison of New and old versions (along the default execution path)
11 0.01047277079593059
```

No performance regression

The 6th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3343
4  The new version is executed by default
5  0.3366
```

```
6 The old version
7 0.327999999999999996
8 Comparison of New and old versions (along symbolic execution path)
9 0.019207317073170818
10 Comparison of New and old versions (along the default execution path)
11 0.026219512195122113
```

No performance regression

The 7th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3424
4 The new version is executed by default
5 0.3363
6 The old version
7 0.333300000000000004
8 Comparison of New and old versions (along symbolic execution path)
9 0.027302730273027122
10 Comparison of New and old versions (along the default execution path)
11 0.009000900090008842
```

No performance regression

The 8th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3395
4  The new version is executed by default
5  0.334900000000000003
6  The old version
7  0.3355
8  Comparison of New and old versions (along symbolic execution path)
9  0.011922503725782425
10 Comparison of New and old versions (along the default execution path)
11 -0.0017883755588673305
```

No performance regression

The 9th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  5.5268999999999995
4  The new version is executed by default
5  5.54590000000000005
6  The old version
7  5.2709
8  Comparison of New and old versions (along symbolic execution path)
9  0.04856855565463191
10 Comparison of New and old versions (along the default execution path)
11 0.05217325314462432
```

There may be performance regression

The 10th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3373000000000000004
4  The new version is executed by default
5  0.3404
6  The old version
7  0.3326
8  Comparison of New and old versions (along symbolic execution path)
9  0.014131088394467941
10 Comparison of New and old versions (along the default execution path)
11 0.02345159350571249
```

No performance regression

The 11th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3438
4  The new version is executed by default
5  0.3411
6  The old version
7  0.3302
8  Comparison of New and old versions (along symbolic execution path)
9  0.04118715929739552
10 Comparison of New and old versions (along the default execution path)
11 0.033010296789824416
```

There may be performance regression

The 12th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  5.4789
4  The new version is executed by default
5  5.3168
6  The old version
7  5.2105
8  Comparison of New and old versions (along symbolic execution path)
9  0.05151137126955199
10 Comparison of New and old versions (along the default execution path)
11 0.02040111313693505
```

There may be performance regression

The 13th symbolic execution

Testcase:SELECT * FROM employee WHERE department_id = 1 AND salary > 6000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.044
4  The new version is executed by default
5  0.0451
6  The old version
7  0.047099999999999996
8  Comparison of New and old versions (along symbolic execution path)
9  -0.06581740976645432
10 Comparison of New and old versions (along the default execution path)
11 -0.042462845010615605
```

No performance regression

The 14th symbolic execution

Testcase:SELECT username FROM users WHERE id = 1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0036
4  The new version is executed by default
5  0.00350000000000000005
6  The old version
7  0.00350000000000000005
8  Comparison of New and old versions (along symbolic execution path)
9  0.028571428571428394
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 15th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3353000000000000004
4  The new version is executed by default
5  0.3414000000000000004
6  The old version
7  0.335999999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0020833333333333104
10 Comparison of New and old versions (along the default execution path)
11 0.016071428571428785
```

No performance regression

The 16th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3409
4  The new version is executed by default
5  0.33399999999999996
6  The old version
7  0.3396
8  Comparison of New and old versions (along symbolic execution path)
9  0.003828032979976348
10 Comparison of New and old versions (along the default execution path)
11 -0.01648998822143713
```

No performance regression

The 17th symbolic execution

Testcase:SELECT * FROM employee WHERE email LIKE '%[@example.com](mailto:example.com)';

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.2806
4  The new version is executed by default
5  0.2807
6  The old version
7  0.27
8  Comparison of New and old versions (along symbolic execution path)
9  0.03925925925925925
10 Comparison of New and old versions (along the default execution path)
11 0.03962962962962958
```

There may be performance regression

The 18th symbolic execution

Testcase:DO ST_AsText(POINT(1.0, 1.0));

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.002
4  The new version is executed by default
5  0.002100000000000000003
6  The old version
7  0.002
8  Comparison of New and old versions (along symbolic execution path)
9  0.0
10 Comparison of New and old versions (along the default execution path)
11 0.05000000000000000013
```

No performance regression

The 19th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3395
4  The new version is executed by default
5  0.3403
6  The old version
7  0.3415
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0058565153733528596
10 Comparison of New and old versions (along the default execution path)
11 -0.0035139092240118134
```

No performance regression

The 20th symbolic execution

Testcase:SELECT id, username, IF(id % 2 = 0, 'even', 'odd') AS id_parity FROM users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.070399999999999999
4  The new version is executed by default
5  0.0708
6  The old version
7  0.0679
8  Comparison of New and old versions (along symbolic execution path)
9  0.03681885125184077
10 Comparison of New and old versions (along the default execution path)
11 0.04270986745213549
```

There may be performance regression

The 21st symbolic execution

Testcase:SELECT user_id, SUM(ABS(id - 10)) AS total_abs_diff FROM posts GROUP BY user_id;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.228799999999999998
4  The new version is executed by default
5  0.236900000000000003
6  The old version
7  0.2338
8  Comparison of New and old versions (along symbolic execution path)
9  -0.021385799828913737
10 Comparison of New and old versions (along the default execution path)
11 0.013259195893926516
```

No performance regression

The 22nd symbolic execution

Testcase:UPDATE employee SET salary = salary * 2 WHERE department_id = 1;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0082
4  The new version is executed by default
5  0.0081
6  The old version
7  0.0086
8  Comparison of New and old versions (along symbolic execution path)
9  -0.04651162790697667
10 Comparison of New and old versions (along the default execution path)
11 -0.058139534883720985
```

No performance regression

The 23rd symbolic execution

Testcase:SELECT id, username, CEIL(id / 2) AS ceil_value FROM users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0943
4  The new version is executed by default
5  0.0962
6  The old version
7  0.094
8  Comparison of New and old versions (along symbolic execution path)
9  0.0031914893617020716
10 Comparison of New and old versions (along the default execution path)
11 0.02340425531914887
```

No performance regression

The 24th symbolic execution

Testcase:SELECT u.username, p.title, CONCAT(p.title, ' by ', u.username) AS full_title FROM users u JOIN posts p ON u.id = p.user_id;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.4422
4  The new version is executed by default
5  0.437100000000000004
6  The old version
7  0.4319
8  Comparison of New and old versions (along symbolic execution path)
9  0.023848112989117796
10 Comparison of New and old versions (along the default execution path)
11 0.012039824033341138
```

No performance regression

The 25th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.33799999999999997
4  The new version is executed by default
5  0.3511
6  The old version
7  0.331400000000000003
8  Comparison of New and old versions (along symbolic execution path)
9  0.019915509957754794
```

```
10 Comparison of New and old versions (along the default execution path)
11 0.05944477972238984
```

No performance regression

The 26th symbolic execution

Testcase:CREATE OR REPLACE VIEW user_post_comments_new AS SELECT u.username, p.title, c.comment FROM users u JOIN posts p ON u.id = p.user_id JOIN comments c ON p.id = c.post_id;

Compilation error occurred, terminated.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.028999999999999998
4 The new version is executed by default
5 0.0444
6 The old version
7 0.0348
8 Comparison of New and old versions (along symbolic execution path)
9 -0.16666666666666666
10 Comparison of New and old versions (along the default execution path)
11 0.2758620689655174
```

No performance regression

The 27th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.33039999999999997
4 The new version is executed by default
5 0.3378
```

```
6 The old version
7 0.335999999999999997
8 Comparison of New and old versions (along symbolic execution path)
9 -0.016666666666666665
10 Comparison of New and old versions (along the default execution path)
11 0.005357142857142928
```

No performance regression

The 28th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3361
4 The new version is executed by default
5 0.3378
6 The old version
7 0.339
8 Comparison of New and old versions (along symbolic execution path)
9 -0.00855457227138647
10 Comparison of New and old versions (along the default execution path)
11 -0.0035398230088496585
```

No performance regression

The 29th symbolic execution

Testcase:SELECT id, username, IF(id % 2 = 0, 'even', 'odd') AS id_parity FROM users;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
```

```
3  0.0769
4  The new version is executed by default
5  0.0669
6  The old version
7  0.0723
8  Comparison of New and old versions (along symbolic execution path)
9  0.0636237897648685
10 Comparison of New and old versions (along the default execution path)
11 -0.07468879668049795
```

No performance regression

The 30th symbolic execution

Testcase:SELECT user_id, COUNT(*) AS post_count, SUM(CASE WHEN LENGTH(content) > 100 THEN 1 ELSE 0 END) AS long_posts FROM posts GROUP BY user_id;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3242
4  The new version is executed by default
5  0.323200000000000004
6  The old version
7  0.3262
8  Comparison of New and old versions (along symbolic execution path)
9  -0.006131207847946051
10 Comparison of New and old versions (along the default execution path)
11 -0.009196811771918906
```

No performance regression

The 31st symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
```

```
2 The new version executes along symbols
3 5.4102
4 The new version is executed by default
5 5.3165999999999999
6 The old version
7 5.1307
8 Comparison of New and old versions (along symbolic execution path)
9 0.05447599742725157
10 Comparison of New and old versions (along the default execution path)
11 0.0362328727074277
```

There may be performance regression

The 32nd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3459
4 The new version is executed by default
5 0.354
6 The old version
7 0.3388
8 Comparison of New and old versions (along symbolic execution path)
9 0.02095631641086185
10 Comparison of New and old versions (along the default execution path)
11 0.04486422668240848
```

There may be performance regression

The 33rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution completed successfully.

代码块
end-to-end diff:

```
2 The new version executes along symbols
3 5.433100000000000005
4 The new version is executed by default
5 5.3062
6 The old version
7 5.2281999999999999
8 Comparison of New and old versions (along symbolic execution path)
9 0.03919130867220099
10 Comparison of New and old versions (along the default execution path)
11 0.01491909261313651
```

There may be performance regression

The 34th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3585
4 The new version is executed by default
5 0.3603
6 The old version
7 0.3386
8 Comparison of New and old versions (along symbolic execution path)
9 0.05877141169521551
10 Comparison of New and old versions (along the default execution path)
11 0.06408741878322503
```

There may be performance regression

The 35th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3408
4  The new version is executed by default
5  0.3466
6  The old version
7  0.3391
8  Comparison of New and old versions (along symbolic execution path)
9  0.005013270421704451
10 Comparison of New and old versions (along the default execution path)
11 0.022117369507519923
```

There may be performance regression

The 36th symbolic execution

Testcase:SELECT user_id, GROUP_CONCAT(title SEPARATOR ', ') AS all_titles FROM posts GROUP BY user_id;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.487699999999999997
4  The new version is executed by default
5  0.4795
6  The old version
7  0.4835
8  Comparison of New and old versions (along symbolic execution path)
9  0.008686659772492207
10 Comparison of New and old versions (along the default execution path)
11 -0.008273009307135478
```

No performance regression

The 37th symbolic execution

Testcase:SET @user_id = 1;

Symbolic execution completed successfully.

代码块

```

1  end-to-end diff:
2  The new version executes along symbols
3  0.0001
4  The new version is executed by default
5  0.0018
6  The old version
7  0.0017000000000000000001
8  Comparison of New and old versions (along symbolic execution path)
9  -0.9411764705882353
10 Comparison of New and old versions (along the default execution path)
11 0.0588235294117646

```

There may be performance regression

Commit 30:

173b16f3bca6e49246f7acd67517eeb2dbbd2d13

Unable to provide test cases

Commit 31:

86ec20189abaf0b07150a0d73c6d6e0f2a3f4780

Unable to provide test cases

Commit 32:

ecb7c9b692811f96cfa54add61012a701c47523d

The 1st symbolic execution

```
Testcase:INSERT INTO users (id, username, email) VALUES (65536, 'user4',
JSON_UNQUOTE(JSON_EXTRACT('{\"email\": \"user4@example.com\"}', '$.email'))) ON DUPLICATE
KEY UPDATE email = VALUES(email);
```

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 2nd symbolic execution

Testcase:INSERT INTO eids(id, eid) VALUES (65536, 72004) ON DUPLICATE KEY UPDATE eid = 72004;

No conditional branches, no need for symbolic execution.

代码块

```

1  end-to-end diff:
2  The new version executes along symbols
3  0.003400000000000000002
4  The new version is executed by default
5  0.0033
6  The old version
7  0.0033
8  Comparison of New and old versions (along symbolic execution path)
9  0.030303030303030384
10 Comparison of New and old versions (along the default execution path)
11 0.0

```

No performance regression

The 3rd symbolic execution

```
Testcase:INSERT INTO users (id, username, email) VALUES (65536, 'user4',
JSON_UNQUOTE(JSON_EXTRACT('{\"email\": \"user4@example.com\"}', '$.email')) ON DUPLICATE
KEY UPDATE email = VALUES(email);
```

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 4th symbolic execution

Testcase:DROP DATABASE IF EXISTS test;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0039
4  The new version is executed by default
5  0.0054
6  The old version
7  0.0056
8  Comparison of New and old versions (along symbolic execution path)
9  -0.3035714285714286
10 Comparison of New and old versions (along the default execution path)
11 -0.03571428571428566
```

No performance regression

The 5th symbolic execution

Testcase:DROP TRIGGER IF EXISTS before_user_insert;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0456
4  The new version is executed by default
5  0.00210000000000000003
6  The old version
7  0.0018
8  Comparison of New and old versions (along symbolic execution path)
9  24.333333333333332
10 Comparison of New and old versions (along the default execution path)
11 0.16666666666666668
```

There may be performance regression

The 6th symbolic execution

Testcase:DROP TRIGGER IF EXISTS before_user_insert;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0021999999999999997
4  The new version is executed by default
5  0.002100000000000000003
6  The old version
7  0.002100000000000000003
8  Comparison of New and old versions (along symbolic execution path)
9  0.047619047619047325
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

There may be performance regression

The 7th symbolic execution

Testcase:INSERT INTO users (id, username, email) VALUES (65536, 'user4',
JSON_UNQUOTE(JSON_EXTRACT('{\"email\": \"[user4@example.com](#)\", '\$.email'}))) ON DUPLICATE
KEY UPDATE email = VALUES(email);

Compilation error occurred, terminated.

代码块

```
1  end-to-end diff:
```

No performance regression

The 8th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3519
4  The new version is executed by default
5  0.3423
6  The old version
7  0.3424
8  Comparison of New and old versions (along symbolic execution path)
9  0.027745327102803766
10 Comparison of New and old versions (along the default execution path)
11 -0.00029205607476632297
```

There may be performance regression

The 9th symbolic execution

Testcase:DELETE FROM users WHERE NOT id < 1000;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.03
4  The new version is executed by default
5  0.0312
6  The old version
7  0.0335
8  Comparison of New and old versions (along symbolic execution path)
9  -0.1044776119402986
10 Comparison of New and old versions (along the default execution path)
11 -0.06865671641791055
```

No performance regression

The 10th symbolic execution

Testcase:CALL GetUserPosts(1);

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0002
4  The new version is executed by default
5  0.0002
6  The old version
7  0.0002
8  Comparison of New and old versions (along symbolic execution path)
9  0.0
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

Commit 33:

ddfebd48f58678ff31427f7f7d971981e2d098ec

Non-C file modifications, therefore no performance regression testing is required.

Commit 34:

18115d392fda06bf90635caaca5801d2ecffdefb

Non-C file modifications, therefore no performance regression testing is required.

Commit 35:

8a95409393b3908fecfa78dcd344b3642683f8c6

Non-C file modifications, therefore no performance regression testing is required.

Commit 36:

f8bc40ef5fd4316e1c84b18b21b83aa6cecb93d4

Unable to provide test cases

Commit 37:

1f85eeeb53ac59c56abc052c97a57ab63115ad29

Unable to provide test cases

Commit 38: efc5d3f84db8277e8b8e01d34ac9dbc7ee9be657

The 1st symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0036
4  The new version is executed by default
5  0.0036999999999999997
6  The old version
7  0.0036999999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  -0.027027027027026983
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 2nd symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0036999999999999997
4  The new version is executed by default
5  0.0038
6  The old version
7  0.0036999999999999997
8  Comparison of New and old versions (along symbolic execution path)
```

```
9 0.0
10 Comparison of New and old versions (along the default execution path)
11 0.0270270270270271
```

No performance regression

Commit 39: c234a312d74afd7b354a51385b19f97c37b9416e

The 1st symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users (id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT NULL, email VARCHAR(100) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.0331
4 The new version is executed by default
5 0.0340
6 The old version
7 0.0340
8 Comparison of New and old versions (along symbolic execution path)
9 -0.0264705882
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 2nd symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
```

No performance regression

The 3rd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1 end-to-end diff:
```

No performance regression

The 4th symbolic execution

Testcase:SELECT DISTINCT u.username, p.title, c.comment FROM users u, posts p, comments c WHERE u.id = p.user_id AND p.id = c.post_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-01') ORDER BY c.comment DESC;

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1 end-to-end diff:
```

No performance regression

Commit 40: 4992aaf9a20bbace98924263f26dea0c1bd97f9c

The 1st symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.2663
4  The new version is executed by default
5  0.2855
6  The old version
7  0.2839
8  Comparison of New and old versions (along symbolic execution path)
9  -0.061993659739344856
10 Comparison of New and old versions (along the default execution path)
11 0.005635787249031315
```

No performance regression